Installation Guidelines

Concealed Installation

Thermoline does not cause any problem when embedded in the wall or floor, because naturally occurring frictional forces prevent the thermal expansion and contraction.

Fastening Technique for Open Installation

Suspended pipelines requires compensation for thermal changes and this can be achieved by proper placement of fixed and sliding clamps in the installation network.

a) Fixed Point:

Fixed clamps help limit the uncontrollable move-ments of the pipelines and divide them into sections. Fixed point spacing must be performed on the basis of pipe diameters. The material used to perform this operation must posses certain characteristics so that it does not damage the external surface of the pipe.

b) Sliding Point:

Sliding clamps allow the axial movement of the pipe without damaging it. On locating a sliding clamp it has to be ensured that movements of the pipeline are not hindered by the fittings or armature installed next to them.

Thermoline Clamp Spaces

Maximum distances of supports of Thermoline PPR - S 3.2 (PN 16) pipe (horizontal pipeline)

Ø pipe (mm)	Spacing Distances (in cm) at Temperature of					
	20°	30°	40°	50°	60°	80°
20	90	80	80	80	70	65
25	95	95	95	90	80	75
32	110	105	105	100	95	80
40	120	120	115	105	100	95
50	135	130	125	120	115	100
63	155	150	145	135	130	115
75	170	165	160	150	145	125
90	180	180	170	165	160	135
110	200	195	190	180	175	155

Maximum distances of supports of Thermoline PPR - S 2.5 (PN 20) pipe (horizontal pipeline)

Ø pipe (mm)	Spacing Distances (in cm) at Temperature of					
	20°	30°	40°	50°	60°	80°
20	95	90	85	85	80	70
25	100	100	100	95	90	85
32	120	115	115	110	100	90
40	130	130	125	120	115	100
50	150	150	140	130	125	110
63	170	160	155	150	145	125
75	185	180	175	160	155	140
90	200	200	185	180	175	150
110	220	215	210	195	190	165

About Dadex

Dadex was established as a public limited company in 1959. We started with the manufacturing of FC roofing systems at our facility in Hyderabad, under the dynamic leadership of our Founder Chairman, Mr. Kassim Dada. Being a forward-looking organization, his son, Mr. Sikander Dada kept up to date with the market, and was instrumental in the diversification with plastic piping systems. Presently we produce various thermoplastic systems such as UPVC, PP-R, PP-RCT, PP, PE.

Headquartered in Karachi, our two manufacturing facilities based in Hyderabad and Sundar Industrial Estate (near Lahore) are equipped with state of the art machinery. Our joint efforts have led Dadex to the ISO 9001, ISO 14001, ISO 45001 and PSQCA certifications.

We aim to set standards in every field we enter. Customer focus, quality and innovation are reflected in every aspect of our business. Offering customized solutions to a cross section of markets for over 66 years, Dadex today is a hallmark of trust and reliability.

We believe that our strength lies in delighting our customers. That is why providing quality products backed by round the clock technical support services are the Dadex promise to all our valued customers.

DADEX Eternit Limited

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UAN: (92-21) 111 000 789, Email: marketing@dadex.com.pk

Dadex Authorised Distributor for Thermoline

Burhani Traders

Plot # 2C, Shop # 5, 27th Tauheed Comm. Street, DHA-V, Karachi.
Tel: 0300-8203417 | E-mail: taherisanitary7@gmail.com

TÜV AUSTRIA ISO 9001 · SGS · SGS · Pakistan Standards

Feb. 2026

Thermoline

HOT & COLD WATER SUPPLY





PP-R

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DADEX

Introduction

Thermoline is a versatile and comprehensive piping solution for transportation and distribution of hot and cold water supply systems, air-conditioning systems, public landscaping and sports facilities etc.

Material

Thermoline piping system is manufactured from Polypropylene Random Copolymer (PPR). PPR material is extensively used in food and medical industries due to its superior characteristics. It exhibits excellent physical and chemical properties at elevated temperatures . These factors make PPR an ideal material for piping systems for transmission of hot and cold water, potable water and other fluids.

Available Range & Standards

Thermoline piping system is available in sizes from 20mm to 110mm (4m length) in PN 16 and PN 20 ratings. The system conforms to the following international standards:

Pipes: DIN 8077-8078
Fittings: DIN 16962

Features & Benefits

- Approved internationally for supply of drinking water.
- Lightweight, easy to transport, and easy to install.
- Easily joined by heat fusion.
- Metal threaded fitting for durability & performance.
- Smooth internal surface for reduced head loss.
- Resistant to abrasion and corrosion.
- Noise free and Resistant to frost.
- Environment friendly and Suitable in seismic areas
- Poor conductor of heat and electricity.

Fields of Application

Thermoline piping system is highly recommended for hot and cold water supply systems in followings:

- Residences, Apartments and Plazas
- Hospitals, Hotels and School buildings.
- Swimming pools.
- Commercial buildings and Offices.

PPR is not recommended for gas applications

Jointing Method

Jointing of Thermoline pipes is carried out by a method called 'Heat Fusion'. This is done by means of a welding machine. The male and female parts of pipes / fittings are joined together to form a heat fusion joint.

Welding Guidelines

- Cut the pipe at the right angle with a cutter.
- Chamfers should be given to the outer ends of the pipe by a knife.
- Mark off the welding depth at the pipe end.
- Always clean the pipe and fittings from burrs, dirt and chips before welding.
- Temperature is adjusted to approximately 260°C in the welding machine.
- Simultaneously, heat the ends of both pipe and fittings as per recommended heating time.
- Push the pipe end into the fitting axially and ensures its alignment of assembly within the specified time period.
- After welding heaters of the machine should be cleaned for the next use.

Available Fittings

Complete range of imported PP-R fittings is available. Special fittings could be made available on demand.



Dimensions of Thermoline Pipes

Nominal Outside Dia (mm)	Minimum Wall Thickness PN 16 (mm)	Minimum Wall Thickness PN 20 (mm)
20	2.8	3.4
25	3.5	4.2
32	4.4	5.4
40	5.5	6.7
50	6.9	8.3
63	8.6	10.5
75	10.3	12.5
90	12.3	15.0
110	15.1	18.3

Nominal OD (mm)	Welding Depth (mm)	Average Heating Time (Sec)	Average Working Time (Sec)	Average Cooling Time (min)
20	14	6	4	2
25	16	7	4	2
32	18	8	6	4
40	20	12	6	4
50	23	18	6	4
63	26	24	8	6
75	28	30	10	8
90	30	40	11	8
110	33	50	12	8